

• Previous videos -

* basic elements of interface $\begin{matrix} \nearrow \text{terms} \\ \rightarrow \text{thms} \end{matrix}$

* Some basic proof methods $\begin{matrix} \nearrow \text{FW} \\ \rightarrow \text{DW} \\ \searrow \text{subst} \end{matrix}$

• This video :-

* Mathematical induction for nat's

• Intuition :-

* Suppose we want to prove a property P holds for all natural numbers n .

E.g. $?x \leq ?x + 3$

* We prove it by induction :-

• we have to prove two things, to show that P holds for all n

$\rightarrow \text{goal}_1$:- P holds for 0 $\xrightarrow{\text{Base case}}$

$\rightarrow \text{goal}_2$:- assuming P holds for n , $\rightarrow$ step case
we can show P holds for $n+1$

• Because we have an algorithm to show that P holds for any given n .

$\rightarrow$ how? $\begin{matrix} \text{goal}_2 \\ \nearrow \\ P_0 \end{matrix} \xrightarrow{\text{goal}_1} P_1 \xrightarrow{?} P_2 \xrightarrow{?} \dots \xrightarrow{?} P_n$

* Reminder: In Isabelle, In the step case "Suc n " denotes $n+1$, only for natural numbers.

• Example :-

$?x + ?y = ?y + ?x$ add - commute

proof by induction on y ,

Base case --- $y = 0$

goal_1 :- $?x + 0 = 0 + ?x$ $\text{lem1 } ?x + 0 = ?x$

1. $?x + 0 = ?x$ by subst lem1 $\text{lem2 } 0 + ?x = ?x$

2. $\dots = 0 + ?x$ by subst lem2

finally $?x + 0 = 0 + ?x$ ✓

Step case

$\xrightarrow{\text{I.H.}}$
if we have the property for n , then we can show it holds for $n+1$

assume

$\boxed{\text{I.H. } ?x + ?y = ?y + ?x}$

goal_2 :- $?x + (\text{Suc } ?y) = (\text{Suc } ?y) + ?x$

1. $?x + (\text{Suc } ?y) = \text{Suc } (?x + ?y)$
by subst lem3

$\text{lem3: } ?a + (\text{Suc } ?b) = \text{Suc } (?a + ?b)$

2. $\dots = (\text{Suc } (?y + ?x))$
by subst I.H.

$\text{lem4: } (\text{Suc } ?a) + ?b = \text{Suc } (?a + ?b)$

3. $\dots = (\text{Suc } ?y) + ?x$
by subst lem4

finally $?x + (\text{Suc } ?y) = (\text{Suc } ?y) + ?x$ ✓

• Notes :-

$\rightarrow$ in Isabelle $\text{Suc } x \approx x + 1$ for nat's